Assignment- 01

Title: K-Means Clustering & K-Nearest Neighbors

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Assignment Set: K-Means Clustering

Assignment: 1

Question: 1

I) clustering: clustering is an unsupervised machine learning technique used to group a set of objects or data points into clusters (group) such that data points in the same cluster are more similar to each other than to those in other clusters.

Key characteristics:

- It is unsupervised because the data has no predefined labels.
- The goal is to discover in inherent structure or patterns in the data.
- Similarity is usually measured using distance matrices.

Types of clustering:

I) partitioning clustering (k-means)
II) Hierarchical clustering
III) Density-based clustering
IV) Model based clustering

In k-means context:

k-means is one of the most popular partitioning algorithms. It divides the dataset into k non-overlapping clusters by minimizing the variance within each cluster.

Example:

- customer segmentation in marketing
- Grouping similar documents or news articles.
- Image compression
- Anomaly detection

ii) centroid: A centroid is the center point (arithmatic mean) of all the data points that belong to a particular cluster.

calculation:

For a cluster with n points, the centroid coordinates are:

$$c_x = \frac{\sum_{i=1}^{n} x_i}{n}, \quad c_y = \frac{\sum_{i=1}^{n} y_i}{n}$$

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Important points:

- centroid are not necessary for actual data points - they are calculated averages.
- In the first iteration, centroid are initialized.
- In every subsequent iteration, centroids are recalculated based on the current cluster members.

Role in K-means:

- Each cluster is represented by one centroid.
- The algorithm tries to find the best position for these centroids so that the total distance between points and their centroids is minimized.

iii) Euclidian Distance: Euclidian distance is the straight line distance between two points in Euclidean space.
It is mostly used distance metric in K-means.

Formula:

For 2D points:

$$d(P,q)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

For n dimention:

$$d(P,q)=\sqrt{\sum_{i=1}^{n}(P_i-q_i)^2}$$

Limitations:

i) Works best when features are on similar scales.

ii) Sensitive to outliers.

iii) Performs poorly in very high dimentions.

Role in k-means:

- It measures how "far" a data point is from a centroid.
- k-means aims to minimize the sum of squared Euclidean distances from points to their cluster centroid (within-cluster sum of squares - called wcss).

IV) Iteration: An iteration in k-means refers to the one complete cycle of the algorithm consisting of two main steps:

- Assignment step: Assign each data point to the nearest centroid.
- update step: Recalculated the centroids based on the new cluster assignments.

Process of one iteration:

i) start with initial centroid.
ii) calculate distance of every point from all centroid.
iii) Assign points to the closest centroid → from clusters.
iv) compute new centroid for each cluster.
v) Repeat until convergence.

Importance of iteration:

- Improves cluster quality.
- Move centroids toward optimal positions.

Role in k-means: k-means is an iterative algorithm. It keeps repeating these steps until the centroids stabilize.

v) convergence: convergence occurs when the algorithm stops making significant changes specifically, when the centroids stop moving between two consecutive iteration.

The algorithm converges when:

- New centroids are exactly the same as the previous centroids.
- cluster assignments remain unchanged.

At this points:

- Further iterations produce the same result.
- Algorithm stops automatically.

Role in k-means:

- The algorithm is trying to minimize the objective function.
- K-Means is guaranteed to converge because WCSS can only decrease or stay the same in each iteration.
- However, it may converge to a local optimum, not necessarily the global best solution.

Question: 02

Difference between supervised learning and unsupervised learning:

Aspect	supervised learning	unsupervised learning
Data type	Labeled data (Input + output)	unlabeled data (only input)
Goal	predict output for new data.	Discover patterns, structure or groupings in data.
Type of problem	prediction (classification and Regression)	Discovery (clustering, Association, Dimentionality Reduction)
Feedback	Model receives direct feedback.	No feedback - model finds pattern by itself.
complexity	Generally easier to evaluate.	Harder to evaluate (no ground truth)
Real-world analogy	student learning with answer key.	student exploring data without guidance.
Training approach	Learn from example with known answer.	Learns by finding similarities/differences.
Examples	KNN classification, decision Tree.	K-Means, PCA

Difference between classification & clustering:

Aspect	classification	clustering
Definition	predicting the class label of new data points.	Grouping similar data points into clusters.
Type of learning	supervised learning.	unsupervised learning
Goal	Learn a mapping from ~~inpt~~ features to predefined classes.	Discover natural groupings or structure in the data.
output	Discrete class labels (Red, Blue, spam, not spam)	cluster IDs or group assignments (cluster-1, 2, 3)
Number of classes	pre-defined (known in advance)	Not known in advance (algorithm decides)
prediction	can predict on completly new unseen data.	Mainly used for analysis; new points can be assigned.
Evaluation Method	Accuracy, precision, Recall, F1-score, confusion Matric.	Internal matrics: silhouette score, WCSS, Davies - Bouldin.
common algorithm	KNN, Decision Tree, Random forest, SVM, Logistic Regression.	K-Means, Hierarchical DBSCAN, Gaussian Matrics Models.

Question: 03

Advantages of K-Means clustering:

i) Simple and Easy understand: The algorithm is conceptually straightforward - assign points to nearest centroid and update centroids. Easy to implement and explain even to non-technical person.

ii) Conceptually Efficient: It is very fast compared to many other clustering algorithms. Time complexity is approximatly $O(n \times k \times i)$, where, n = number of data points, k = number of clusters, i = number of iterations. Suitable ~~for~~ large datasets.

iii) Scales well to High Dimensions: Works reasonably well when the number of features (dimensions) is not extremely high.

iv) Guaranteed convergence: The algorithm always convergence because the within-cluster sum of square (WCSS) decreases or stays the same in every iteration.

v) Produces Tight, spherical clusters: Works very well when clusters are roughly spherical, compact and of similar size.

vi) Easy to interpret and implement: only a few parameters (mainly k and distance metric). Many libraries provide excellent implementations.

vii) Versatile Application: widely used in customer segmentation, image compression, docoment clustering, anomaly detection etc.

used in various field:

- Image processing.
- Medical diagnosis.
- Docoment clustering.
- Recomendation systems.

Disadvantages of K-Means Algorithm:

i) Must specify K in Advance: You need to decide the number of cluster (K) before running the algorithm. choosing the wrong K leads to poor results.

ii) sensitive to Initial centroid selection: Different random initial centroids can produce very different final clusters. It can get stuck in local optima.

iii) sensitive to outlier: outliers can significantly pull the centroids away from the center of the actual clusters, distorting the results.

iv) Assumes spherical and Equally sized clusters: performs poorly if clusters have irregular shapes, different sizes, or different densities.

v) sensitive to Feature scaling: If features have different scale, distance calculations become biased. Feature normalization/ standardization is mandatory.

vi) Not suitable for Non-convex clusters: can't handle complex, non-linear cluster shapes. Algorithms like DBSCAN perform better here.

vii) Hard to Evaluate: Since there are no ground-truth labels, evaluating the quality of clusters is subjective and difficult.

viii) Curse of Dimensionality: performance degrades significantly when the number of dimensions is very high.

Real-world Example of Advantages:
A retail company wants to segment customers based on purchasing behavior. K-Means can quickly group millions of customers into 5-10 segments efficiently.

Real world Example of Disadvantages:
If you apply K-Means on customer data where one cluster is very dense and another is sparse, or if there are extreme outliers (VIP customers), the result can be misleading.

Assignment: 2

Given:

point	x	y
A	1	1
B	2	1
C	4	3
D	5	4

Number of clusters, $k = 2$

Initial centroids:

$$c_1 = (1,1)$$
$$c_2 = (5,4)$$

(a) Calculate Euclidean Distance:

Formula:

$$\text{Distance} = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

point $A(1,1)$:

Distance to $c_1(1,1)$: $\sqrt{(0-0)^2 + (0-0)^2}$
$= \sqrt{0} = 0$

Distance to $c_2(5,4) = \sqrt{(5-1)^2 + (4-1)^2}$
$= \sqrt{16+9} = \sqrt{25} = 5$

point $(2,1)$:

Distance to $c_1(1,1) = \sqrt{(2-1)^2 + (1-1)^2}$
$= \sqrt{1^2+0^2} = 1$

Distance to $c_2(5,4) = \sqrt{(5-2)^2+(4-1)^2}$

$= \sqrt{9+9} = \sqrt{18} \approx 4.24$

point $c(4,3)$:

Distance to $c_1(1,1) = \sqrt{(4-1)^2+(3-1)^2}$

$= \sqrt{9+4} = \sqrt{13} \approx 3.606$

Distance to $c_2(5,4) = \sqrt{(5-4)^2+(4-3)^2}$

$= \sqrt{1+1} = \sqrt{2} \approx 1.414$

point $D(5,4)$:

Distance to $c_1(1,1) = \sqrt{(5-1)^2+(4-1)^2}$

$= \sqrt{16+9} = \sqrt{25} = 5$

Distance to $c_2(5,4) = \sqrt{(5-5)^2+(4-4)^2}$

$= \sqrt{0^2+0^2} = 0.$

(b) We assign each point to the cluster whose centroid is closest (minimum distance).

point	Distance to c_1	Distance to c_2	Nearest centroid	Assigned cluster
A	0	5	c_1	cluster 1
B	1	4.24	c_1	Cluster 1
C	3.606	1.414	c_2	cluster 2
D	5	0	c_2	cluster 2

Final assignment after step 1:

cluster 1: points A, B.

cluster 2: points C, D.

ⓒ compute new centroid:

New centroid = Average (Mean) of all points in that cluster.

New centroid C_1 (cluster 1):

points : A(1,1) and B(2,1)

$\therefore$ New $X = \frac{1+2}{2} = 1.5$

$\therefore$ New $Y = \frac{1+1}{2} = 1$

$\therefore$ New $C_1 = (1.5, 1)$.

New centroid C_2 (cluster 2):

points: C(4,3) and D(5,4)

$\therefore$ New $X = \frac{4+5}{2} = 4.5$

$\therefore$ New $Y = \frac{3+4}{2} = 3.5$

$\therefore$ New, $C_2 = (4.5, 3.5)$

(d) step by step Analysis:

After completing the first iteration, we have:

New centroid:

$$c_1 = (1.5, 1)$$
$$c_2 = (4.5, 3.5)$$

To check convergence, we must perform the second iteration and see if the centroids change again.

second iteration:

step: I

point $A(1,1)$:

$$\text{TO } c_1 (1.5, 1) = \sqrt{(1.5-1)^2 + (1-1)^2}$$
$$= 0.5$$

$$\text{TO } c_2 (4.5, 3.5) = \sqrt{(4.5-1)^2 + (3.5-1)^2}$$
$$= \sqrt{12.25 + 6.25} = \sqrt{18.5} \approx 4.30$$

point $B(2,1)$:

$$\text{TO } c_1 (1.5, 1) = \sqrt{(2-1.5)^2 + (1-1)^2}$$
$$= 0.5$$

$$\text{TO } c_2 (4.5, 3.5) = \sqrt{(4.5-2)^2 + (3.5-1)^2}$$
$$= \sqrt{6.25 + 6.25} = \sqrt{12.5} \approx 3.54$$

point (4,3):

TO $c_1(1.5,1) = \sqrt{(4-1.5)^2+(3-1)^2}$

$= \sqrt{6.25+4} = \sqrt{10.25}$

≈ 3.20

TO $c_2(4.5,3.5) = \sqrt{(4.5-4)^2+(3.5-3)^2}$

$= \sqrt{0.25+0.25} = \sqrt{0.5} \approx 0.707$

point D(5,4):

TO $c_1(1.5,1) = \sqrt{(5-1.5)^2+(4-1)^2}$

$= \sqrt{12.25+9}$

$= \sqrt{21.25} \approx 4.61$

TO $c_2(4.5,3.5) = \sqrt{(4.5-5)^2+(3.5-4)^2}$

$= \sqrt{0.25+0.25}$

$= \sqrt{0.5} \approx 0.707$

step: II Assign points to nearest cluster.

point	DIST. TO C1	DIST to C2	Nearest	cluster
A	0.5	4.30	C1	1
B	0.5	3.54	C1	1
C	3.20	0.707	C2	2
D	4.61	0.707	C2	2

cluster assignment remains the same:

Cluster 1: A, B

cluster 2: C, D

Step: III compute new centroids

New $c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1.5, 1)$

New $c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4.5, 3.5)$

Final conclusion for convergence:

old centroid (1.5, 1) and (4.5, 3.5)

New centroid (1.5, 1) and (4.5, 3.5)

The centroid did not change between the second iteration and the previous one.

(d) Elbow Method: Elbow Method is one of the most popular and widely used technique for determining the optimal number of clusters (k) in the k-means algorithm.

Procedure:

Step: I Run K Means multiple times

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Execute the K-Means algorithm for a range of K values.

Step:II calculate WCSS for each K

After the algorithm converges for each K, compute the WCSS.

Step:III plot the graph

X-axis : Number of clusters (K)

Y-axis : WCSS value.

Step:IV Identify the Elbow point

Look at the point, plot and find the point where the curve bends sharply - this look like an elbow.

Step:V choose optimal K

The value of K at the elbow point is considered the optimal number of clusters.

Mathematical formula:

For a given K,

$$WCSS(K) = \sum_{i=1}^{k} \sum_{x \in C_i} ||x_i - \mu_i||^2$$

Where:

C_i = cluster i

μ_i = centroid of cluster i

$\|x-\mu_i\|$ = Euclidean distance

The optimal k is chosen at the "elbow" point of the curve, where increasing k further yields diminishing returns in reducing WCSS.

Assignment Set: K-Nearest Neighbors (KNN)

K-Nearest Neighbour

Assignment: 1

Question: 01

1) K-Nearest Neighbour: KNN is a popular supervised machine learning algorithm used for both classification and regression problems.

It is a non-parametric and instance-based algorithm that does not make any assumptions about the underlying data distribution.

How it works: When a data points comes, KNN calculates the distance between this points and all the points in the training dataset, identifies the K-closest points, and makes a prediction based on them. For classification, it uses majority voting, and for regression it takes the average of the neighbors.

It is also called a lazy learner because it delays all computation until predection time and stores the entire training data.

ii) Instance based learning: Instance-based learning (memory-based learning) or lazy learning is a machine learning paradigm where the model does not build an explicit generalized model during the training phase. Instead, it simply stores all the training instances in memory.

When a prediction is needed for a new instance, the algorithm compares it with the stored instances using a similarity or distance measure and uses the most popular or similar instances to make the decision. KNN is the most well-known example of instance-based learning. This approach is very flexible but can be computationally expensive during prediction.

iii) Distance metric: A distance metric or distance function is a mathematical formula used to measure the similarity or dissimimilarity between two data points in a multi-dimensional space.

It plays a crucial role in KNN because the algorithm completely depends on calculating distances to find nearest neighbors. The most commonly used distance metric is Euclidean Distance. Other popular distance metric include:

- Manhattan Distance
- Minkowski Distance
- cosine similarity

coosing the right distance matric significantly affects the performance of the KNN algorithm.

iv) classification: classification is a type of supervised learning task in ML where the goal is to predict the discrete category or class label of a given input data point. The output variable is categorical.

Example:

- predicting whether an email is span or Not spam.
- classification a tumor as Malignant or Benign.
- Identifying a handwritten digits (0 to 9)

common classification algorithms include KNN, Decision Trees, Random Forest, SVM, and Logistic Regression.

v) Regression: Regression is a type of supervised learning task where the goal is to predict a continuous numerical value for a given input. unlike classification, the output is not a category but a quantity.

Example:

- predicting house price based on size, location etc.
- predicting student exam score based

on study hour.
- Forecasting temperature based on weather features.

In KNN Regression, the prediction is made by taking the average value of the k nearest neighbors.

Question:03

Advantages of KNN Algorithm:

I) Simple and Easy to understand: KNN is one of the simplest ML algorithms. It is easy to implement and intuitive. The logic behind it is very easy to explain and visualize, making it excellent for beginners and educational purpose.

II) No training phase (Lazy Learner): KNN is a lazy learning algorithm. It does not build any model during the training phase. It simply stores the training data. This makes the training very fast - almost instantaneous.

iii) versatile: KNN can be used for both classification (prediction categories) and Regression (prediction continuous values) problems.

iv) No parametric: It makes no assumptions about the underlying data distribution. This makes it useful when the data does not flow a specific pattern (lineare or normal distribution)

v) Adaptable to new data: Since it stores all training data, adding new data point is very easy. you don't need to retain the entire model.

vi) Works well with small Datasets: KNN performs quite well when the dataset is relatively small and the number of features is not too high.

viii) High interpretability: you can easily explain why a particular prediction was made by showing the nearest neighbors used for the decision.

Disadvantages of KNN Algorithm:

i) computationally Expensive at prediction Time: For every new test point, KNN must calculate the distance to all training points. This makes prediction very slow, especially on large datasets.

ii) High memory Requirement: Since KNN stores the entire training datasets in memory, it requires a lot of storage space for large datasets.

iii) sensitive to Irrelevant and Redundant Features: KNN uses distance calculations. If there are irrelevant or highly correlated features, they can distort the distance measurement and reduce accuracy. Feature scaling is mandatory.

IV) Curse of Dimensionality: As the number of features (dimensions) increases, the distance between points becomes less meaningful. performance drops significantly in high-dimensional data.

V) sensitive to Noise and outliers: Noisy data or outliers can havily influence the prediction because they may become one of the nearest neighbors.

VI) choice of K is critical: The performance of KNN havily depends on the value of K.

- Small K → Sensitive to noice (overfitting)
- Large K → Overly generalized (underfitting)

VII) Does not work well with Imbalanced data: In classification, if one class has significantly more samples, KNN tends to favor the majority class.

Question: 02

Aspect	KNN classification	KNN Regression
Definition	predicts the class label of a new data point.	Predicts a continuous numerical value for a new data point
Type of task	classification (categorical output)	Regression (continuous output)
prediction Method	Majority voting - The class with the most votes among k neighbors wins.	Average (Mean) - Takes the average value of k nearest neighbors
output	Discrete class label (Yes/No, spam/Ham)	continuous value (45.6, 32000, ...)
Example	predict a tumor is Malignant or Benign.	predict the price of a house.
Evaluation Metrics	Accuracy, precision, Recall, F1-score, confusion Matrix.	Mean square Error, Root Mean squeare Error, Mean Absolute Error.
Decision boundary	create clear class boundary.	predicts smooth numerical values.
use case.	Image recognition, Email spam detection, Diseases diagnosis.	House price prediction, stock price forecasting salary prediction.

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Assignment: 02

Given data

point	X	Y	class
A	1	1	Red
B	2	2	Red
C	4	4	Blue
D	5	5	Blue

New test point:

$$P = (3,3)$$

Q: 1

Euclidean Distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

i) Distance between P(3,3) and A(1,1):

$$d(A,P) = \sqrt{(3-1)^2 + (3-1)^2}$$
$$= \sqrt{4+4} = \sqrt{8} \approx 2.828$$

ii) $d(B,P) = \sqrt{(3-2)^2 + (3-2)^2}$
$$= \sqrt{1+1} = \sqrt{2} \approx 1.414$$

iii) $d(C,P) = \sqrt{(3-4)^2 + (3-4)^2}$
$$= \sqrt{1+1} = \sqrt{2} \approx 1.414$$

iv) $d(D,P) = \sqrt{(3-5)^2 + (3-5)^2}$

$= \sqrt{4+4} = \sqrt{8} \approx 2.828$

Q: 2

we sort the points based on increasing order of distance:

B(2,2) → 1.414 (Red)
C(4,4) → 1.414 (Blue)
A(1,1) → 2.828 (Red)

D is also 2.828, but we take the 3rd closet.

∴ 3 nearest neighbors: B, C, A.

Q: 3

Now, we look at the classes of 3 nearest neighbors:

B → Red
C → Blue
A → Red

Voting Result:

Red = 2 votes
Blue = 1 vote

Final prediction:

The class of point P(3, 3) is Red (by majority voting).

Assignment: 03

Question: 3

Step-by-step procedure for the KNN algorithm:

Step: I Load the training dataset

Import the labeled training data (features + class label).

Step: II choose the value of K

select the number of nearest neighbors (K). It is usually an odd number to avoid ties (e.g. 3, 5, 7..)

Step: III process the data

- Normalize or standardize the features
- Handle missing values if any.

Step: IV For a new Test point (P)

- calculate the distance between the test point p and every point in the training dataset.
- store these distances along with the corresponding class labels.

Step: V sort the Distances

sort all the training points in ascending order of their distance from p.

Step: VI select k nearest neighbors

pick the top k points with the smallest distance

Step: VII perform majority voting

count the class labels of these k nearest neighbors. The class that appears most frequently (highest votes) is chosen as the predicted class for the test point P.

Step: VIII output the prediction

Assign the majority class to the new point p.

Step! IX Evaluate the Model

use metrics like Accuracy, precision, Recall or F1-score on the test data.

Question:02

The parameter K is one of the most important hyperparameters in the KNN algorithm. It represents the number of nearest neighbors considered when making a prediction.

Role of K:

- controls the Bias variance Tradeoff.
  - small K: (K = 1 or 3)
    → Low bias, high varience
    → The model is more flexible and captures local patterns.
    → However, it is very sensitive to noise and outliers.
  - large K: (K = 10 or more)
    → High bias, Low varience.

→ The model become smoother and more stable.

→ But it may ignore local patterns and underfit the data.

- Decision Boundary
  - A small k creates a complex, wiggly decision boundary.
  - A large k creates a smoother, more linear-like decision boundary.

- Effect on prediction
  - In classification: k determines how many neighbors vote for the final class.
  - In Regression: k determines how many neighbors are averaged for the final value.

- choose optimal k:
  - use cross validation
  - common rule of thumb: $k=\sqrt{n}$ (where n is the number of training samples)
  - k should usually be an odd number to prevent voting ties.

Question: 3

I) If K = 1

- The new point is assigned the class of the single nearest neighbor.
- Effect: very high sensitivity to noise and outliers.
- If the closest point is noisy or mislabeled, the prediction will be wrong.
- Results in high variance, and over-fitting.
- Decision boundary becomes very complex and jagged.

II) If K is very large:

- The algorithm considers almost all training points for every prediction.
- Effect: The model becomes too simple.
- It will usually predict the majority class of the entire dataset.
- Leads to the high bias and under-fitting.

- Loses the ability to capture local patterns in the data.

iii) IF the dataset contains noise:

- Noisy points can become part of the nearest neighbors.
- Effect: significantly degrades the performance of KNN.
- Wrong neighbors lead to incorrect predictions.
- KNN becomes unrelible because it gives equal importance to all neighbors.

Solutions:

i) clean the data (remove outliers).
ii) use a large K (to reduce noise impact).
iii) Apply distance-weighted voting (closer neighbors get more weight).
iv) use feature scaling and dimensionality reduction.